** Project Day - 2 **

```
Name: Ankit Kumar
Batch: Data Analyst 22
Date: 25/07/2023
```

** Spotify Youtube Data Analysis **

```
!pip install -q -U kaggle
!mkdir ~/.kaggle
!cp kaggle.json ~/.kaggle/
!ls ~/.kaggle
!chmod 600 ~/.kaggle/kaggle.json
!kaggle datasets download -d rankirsh/data-7
```

```
Downloading data-7.zip to /content
  0%|                                               |  0.00B/  0.00B |
100%|#################################################|  0.00B/  0.00B |
```

	Unnamed: 0	Artist	Int_popularity	Track	Album	Album_type	Id	Danceability	Energy	Key	...	Id_artists	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	7wGoVQkr9jvGgqp3proPZ3	0.678	0.732	6	...	3AA28KZvwAJcZKsA5MW7CT	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3
1	1	Gorillaz	62	Tarantula	Cracker Island	single	2dHHgzDwk4BJ8ibKjooW0z	0.665	0.907	8	...	3AA28KZvwAJcZKsA5MW7CT	Gorillaz - Tarantula	Gorillaz	548087	15630	475	Tarantula (feat. Beck) [Remix] [Ex...	False	True	171773325	https://open.spotify.com/track/2dHHgzDwk4BJ8ibKjooW0z
2	2	Gorillaz	67	Skyzoo (feat. Posdnuos)	Skyzoo (feat. Posdnuos)	single	3dx71tVnHX8QCzCZyW6OOO	0.923	0.683	0	...	3AA28KZvwAJcZKsA5MW7CT	Gorillaz - Skyzoo (feat. Posdnuos)	Gorillaz	1064964	29966	1000	Skyzoo (feat. Posdnuos) [Remix] [E...	False	True	163021779	https://open.spotify.com/track/3dx71tVnHX8QCzCZyW6OOO
3	3	Gorillaz	67	Silent Running (feat. Bo...	Silent Running (feat. Bo...	single	6Ob7ZzAZ8eIsiNcZ9Ldtqn	0.449	0.732	7	...	3AA28KZvwAJcZKsA5MW7CT	Gorillaz - Silent Running (feat. Bo...	Gorillaz	1064964	29966	1000	Silent Running (feat. Bo...	False	True	163021779	https://open.spotify.com/track/6Ob7ZzAZ8eIsiNcZ9Ldtqn
4	4	Gorillaz	67	Desole	Cracker Island	single	2Ldq1Zqzzzzzzzzzzzzzzz	0.703	0.762	2	...	3AA28KZvwAJcZKsA5MW7CT	Gorillaz - Desole	Gorillaz	1064964	29966	1000	Desole [Remix] [Ex...	False	True	163021779	https://open.spotify.com/track/2Ldq1Zqzzzzzzzzzzzzzz

```
[5 rows x 20 cols]
```

```
data.info()
```

```
<class pandas.core.frame.DataFrame>
RangeIndex: 20718 entries, 0 to 20717
Data columns (total 20 cols):
 #   Unnamed: 0  20718 non-null  int64
 #   Artist      20718 non-null  object
 #   Int_popular 20718 non-null  int64
 #   Track       20718 non-null  object
 #   Album       20718 non-null  object
 #   Album_type  20718 non-null  object
 #   Id          20718 non-null  object
 #   Danceability 20718 non-null  float64
 #   Energy      20718 non-null  float64
 #   Key         20718 non-null  int64
 #   Loudness    20718 non-null  float64
 #   Speechiness 20718 non-null  float64
 #   Acoustiness 20718 non-null  float64
 #   Instrumental 20718 non-null  float64
 #   Liveness    20718 non-null  float64
 #   Valence     20718 non-null  float64
 #   Tempo       20718 non-null  float64
 #   Id_artists  20718 non-null  object
 #   Title       20718 non-null  object
 #   Channel     20718 non-null  object
dtypes: int64(4), float64(12), object(4)
memory usage: 3.2 MB
```

```
data.drop('Unnamed: 0', inplace=True)
```

```
data.columns
```

```
Index(['Artist', 'Int_popular', 'Track', 'Album', 'Album_type',
       'Id', 'Danceability', 'Energy', 'Key', 'Loudness', 'Speechiness',
       'Acoustiness', 'Instrumental', 'Liveness', 'Valence', 'Tempo',
       'Id_artists', 'Title', 'Channel'],
      dtype='object')
```

```
data.shape
```

```
(20718, 19)
```

```
data.dtypes
```

```
Artist          object
Int_popular      int64
Track           object
Album           object
Album_type      object
Id              object
Danceability   float64
Energy         float64
Key              int64
Loudness       float64
Speechiness    float64
Acoustiness    float64
Instrumental   float64
Liveness       float64
Valence        float64
Tempo          float64
Id_artists      object
Title           object
Channel         object
dtype: object
```

```
data.isnull().sum()
```

```
Artist          0
Int_popular     0
Track           0
Album           0
Album_type      0
Id              0
Danceability    0
Energy          0
Key             0
Loudness        0
Speechiness     0
Acoustiness     0
Instrumental    0
Liveness        0
Valence         0
Tempo           0
Id_artists      0
Title           0
Channel         0
dtype: int64
```

```
data.duplicated().sum()
```

```
0
```

```
data.drop(['Id', 'Id_artists'], inplace=True)
```

```
data.shape
```

```
(20718, 17)
```

```
data.dtypes
```

```
Artist          object
Int_popular      int64
Track           object
Album           object
Album_type      object
Danceability   float64
Energy         float64
Key              int64
Loudness       float64
Speechiness    float64
Acoustiness    float64
Instrumental   float64
Liveness       float64
Valence        float64
Tempo          float64
Title           object
Channel         object
dtype: object
```

```
data.describe()
```

	Int_popular	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo
count	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000	20718.000000
mean	64.000000	0.655000	0.653000	5.340000	-7.000000	0.091000	0.239000	0.016000	0.190000	0.513000	122.000000
std	16.000000	0.145000	0.151000	3.500000	2.000000	0.091000	0.207000	0.090000	0.154000	0.233000	23.000000
min	0.000000	0.122000	0.014000	0.000000	-46.000000	0.023000	0.000000	0.000000	0.012000	0.035000	34.000000
25%	62.000000	0.557000	0.559000	2.000000	-8.000000	0.036000	0.055000	0.000000	0.098000	0.331000	104.000000
50%	67.000000	0.670000	0.670000	5.000000	-6.000000	0.048000	0.178000	0.000000	0.127000	0.513000	120.000000
75%	73.000000	0.763000	0.763000	8.000000	-5.000000	0.091000	0.385000	0.000000	0.244000	0.696000	137.000000
max	100.000000	0.975000	0.998000	11.000000	0.000000	0.964000	0.996000	0.910000	0.996000	0.975000	217.000000

```
data.describe(include='object')
```

	Artist	Track	Album	Album_type	Title	Channel
count	20718	20718	20718	20718	20718	20718
unique	1	1	1	1	1	1
top	Gorillaz	Cracker Island (feat. Gu...	Cracker Island	single	Gorillaz - Cracker Island (feat. Gu...	Gorillaz
freq	20718	20718	20718	20718	20718	20718

Q.1) Top 10 Artists with the Highest views on YouTube?

```
data.sort_values('Views', ascending=False).head(10)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

```
data.groupby('Artist').sum().head(1).T.sort
```

```
Artist
Gorillaz    1.0
dtype: int64
Artist
Gorillaz    1.0
dtype: int64
```

```
data.groupby('Artist').sum().sort_values(ascending=False)
```

```
Artist
Gorillaz    20718.0
dtype: int64
```

```
data.groupby('Artist').sum().head(1).T
```

Artist	Gorillaz
Int_popular	1327000.0
Danceability	13570.000000
Energy	13529.000000
Key	110600.0
Loudness	-145000.000000
Speechiness	1886.000000
Acoustiness	4950.000000
Instrumental	332.000000
Liveness	3936.000000
Valence	10627.000000
Tempo	2527000.000000
Title	Gorillaz - Cracker Island (feat. Gu...
Channel	Gorillaz

```
data.sort_values('Views', ascending=False).head(1).T
```

	0
Artist	Gorillaz
Int_popular	79
Track	Cracker Island (feat. Gu...
Album	Cracker Island
Album_type	single
Danceability	0.678
Energy	0.732
Key	6
Loudness	-7.062
Speechiness	0.0363
Acoustiness	0.0408
Instrumental	0.000097
Liveness	0.0846
Valence	0.805
Tempo	105.889
Title	Gorillaz - Cracker Island (feat. Gu...
Channel	Gorillaz
Views	19293391
Likes	548913
Comments	12444
Description	Stylophone by Jeremy Wheatley. Ma...
Licensed	True
Official_video	True
Stream	232171866
URL_spotify	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

Q.2) Top 10 Tracks with the Highest Streams on Spotify?

```
data.sort_values('Stream', ascending=False).head(10)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.3)What are the most common Album Type on Spotify? How many tracks belong to each album type?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
data.Album_type.value_counts
```

```
<bound_method_wrapper.__call__ of Album_type
single    20718
Name: count, dtype: int64>
```

```
data.Album_type.value_counts()
```

```
Album_type
single    20718
Name: count, dtype: int64
```

```
data.Album_type.value_counts().plot.pie
```

```
<bound_method_DataFrameGroupBy.pie of <pandas.core.frame.DataFrame object at 0x7f1c0b0a0a00>>
```

```
data.Album_type.value_counts().plot.pie(autopct='%.2f%%')
```

```
<Axes: ylabel='count'>
```

```
data.Album_type.value_counts.plot.pie(autopct='%.2f%%',"red","green","yellow",explode=(0,0,0),shadow=1)
```

```
<bound_method_wrapper.__call__ of <pandas.core.plotting._matplotlib.tools.PlotAccessor object at 0x7f1c0b0a0a00>>
```

```
%matplotlib inline
%matplotlib inline
```

Q.4) How do the Average Views,Likes and Comments are compared between different Album Types?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

```
group.plot('Views','Likes','Comments','Album')
```

```
Views  Likes  Comments  Album
0      19293391  548913     12444  Cracker Island
```

```
%matplotlib inline
```

```
group.corr.head
```

```
<bound_method>
```

```
group.corr.head()
```

	Views	Likes	Comments
Views	1.000000	0.000000	0.000000
Likes	0.000000	1.000000	0.000000
Comments	0.000000	0.000000	1.000000

```
group.Album_type.value_counts
```

```
Album_type
single    20718
Name: count, dtype: int64
```

```
group.Album_type.value_counts()
```

```
Album_type
single    20718
Name: count, dtype: int64
```

```
group.Album_type.value_counts().plot.bar
```

```
<bound_method_DataFrameGroupBy.bar of <pandas.core.frame.DataFrame object at 0x7f1c0b0a0a00>>
```

```
group.Album_type.value_counts().plot.bar(stacked=True,legend=True,figsize=(12,6))
```

```
<Axes: xlabel='Album_type'>
```

Q.5) Top 5 Youtube Channels based on the Views?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

```
group.sort_values('Views',ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

```
group.Channel.value_counts()
```

```
Channel
Gorillaz    20718
Name: count, dtype: int64
```

```
group.Channel.value_counts().head(5).T.plot.barh
```

```
<bound_method_wrapper.__call__ of <pandas.core.plotting._matplotlib.tools.PlotAccessor object at 0x7f1c0b0a0a00>>
```

Q.6)The Top Most track Based on Views?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.7) Which Top 3 Tracks have the highest Likes to View ratio on YouTube?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

```
group.sort_values('Views',ascending=False).head(3)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.8) Which Top 3 Tracks have the highest Views to Likes ratio on YouTube?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.9) Top albums having the Tracks with Maximum Danceability?

```
data.sort_values('Danceability',ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.10) Top albums having the Tracks with Maximum Danceability ?

```
data.sort_values('Danceability',ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.11) Top 10 Artists with the Highest views on YouTube?

```
data.sort_values('Views', ascending=False).head(10)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.12) Top 10 Tracks with the Highest Streams on Spotify?

```
data.sort_values('Stream', ascending=False).head(10)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

Q.13) Which is the most common Album Type?

```
data.Album_type.value_counts
```

```
<bound_method_wrapper.__call__ of Album_type
single    20718
Name: count, dtype: int64>
```

```
data.Album_type.value_counts()
```

```
Album_type
single    20718
Name: count, dtype: int64
```

```
data.Album_type.value_counts().plot.pie
```

```
<bound_method_DataFrameGroupBy.pie of <pandas.core.frame.DataFrame object at 0x7f1c0b0a0a00>>
```

```
data.Album_type.value_counts().plot.pie(autopct='%.2f%%')
```

```
<Axes: ylabel='count'>
```

Q.14) Is there any correlation between Views, Likes, Comments and Stream?

```
data.sort_values('Views', ascending=False).head(1)
```

	Artist	Int_popular	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechiness	Acoustiness	Instrumental	Liveness	Valence	Tempo	Title	Channel	Views	Likes	Comments	Description	Licensed	Official_video	Stream	URL_spotify
0	Gorillaz	79	Cracker Island (feat. Gu...	Cracker Island	single	0.678	0.732	6	-7.062	0.0363	0.0408	0.000097	0.0846	0.805	105.889	Gorillaz - Cracker Island (feat. Gu...	Gorillaz	19293391	548913	12444	Stylophone by Jeremy Wheatley. Ma...	True	True	232171866	https://open.spotify.com/track/7wGoVQkr9jvGgqp3proPZ3

```
[5 rows x 19 cols]
```

```
group.corr()
```

	Views	Likes	Comments	Stream
Views	1.000000	0.000000	0.000000	0.000000
Likes	0.000000	1.000000	0.000000	0.000000
Comments	0.000000	0.000000	1.000000	0.000000
Stream	0.000000	0.000000	0.000000	1.000000

```
group.corr().plot.pie(autopct='%.2f%%')
```

```
<Axes: ylabel='Views', 'Likes', 'Comments', 'Stream'>
```

```
group.corr().plot.bar(stacked=True,legend=True,figsize=(12,6))
```

```
<Axes: >
```



```
group.corr().plot.barh(stacked=True,legend=False,figsize=(12,6))
```

```
<Axes: >
```



```
group.corr().plot.kde(stacked=True,legend=False,figsize=(12,6))
```

```
<Axes: >
```



```
group.corr().plot.heatmap(stacked=True,legend=False,figsize=(12,6))
```

```
<Axes: >
```